SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA, U.P.

An Internship Report on

URL SHORTENER APPLICATION

Python Developer Internship Report

Submitted in partial fulfilment of the requirement for the degree of
B.Tech (Computer Science and Engineering)

Name of Student:	Prince Maurya
Roll Number:	25SCS1003001538
Programme:	B.Tech CSE
Semester / Section:	3rd Semester / Section 2CSE7
Batch:	2025-2029
Industry Partners:	upSkill Campus & UniConverge Technologies Pvt. Ltd. (UCT)
Collaboration:	AICTE (All India Council for Technical Education)
Certificate ID:	USC838112TIA
Internship Period:	12-08-2026 to 02-09-2026

Academic Session: 2025-2029

CANDIDATE'S DECLARATION

I, **Prince Maurya**, hereby declare that this internship report titled **"Python Developer Internship - URL Shortener Application"** has been prepared by me as part of my academic internship requirement for the B.Tech Computer Science and Engineering programme at IILM University, Greater Noida.

This report presents my original internship experience, learning activities, technical implementation, testing, and project work carried out during the Python developer internship with upSkill Campus and UniConverge Technologies Pvt. Ltd. (UCT) in collaboration with AICTE.

I understand that I am responsible for the accuracy and authenticity of the material submitted in this report.

Student Name	Prince Maurya
Roll / Enrollment No.	25SCS1003001538
Programme	B.Tech CSE
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Batch	2025-2029
Date	September 2, 2026

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Signature of Student: Prince Maurya

ACKNOWLEDGEMENT

I express my sincere gratitude to **IILM University, Greater Noida**, and the **School of Computer Science and Engineering** for providing the academic framework and opportunity to complete my industrial internship as part of the B.Tech programme.

I am deeply thankful to **upSkill Campus** and its industry partner **UniConverge Technologies Pvt. Ltd. (UCT)**, along with the **All India Council for Technical Education (AICTE)**, for providing this Python internship opportunity and practical exposure through hands-on training and industry-oriented project work.

I extend my gratitude to **Kaushlendra Singh Sisodia** (Chief Mentor at The IoT Academy & Director at UniConverge Technologies) and the mentorship team for guiding us through structured weekly milestones. Finally, I thank my faculty coordinators, family, and peers who supported me throughout this learning journey.

Prince Maurya
Roll No.: 25SCS1003001538
Section: 2CSE7 | B.Tech CSE

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1. INTERNSHIP OVERVIEW

This report documents the industrial Python developer internship completed by **Prince Maurya**. The internship program was offered by **upSkill Campus** in collaboration with industry partner **UniConverge Technologies Pvt. Ltd. (UCT)** and supported by the **All India Council for Technical Education (AICTE)**.

As per official records, the internship commenced on **12-08-2026** and was successfully completed on **02-09-2026** under Certificate ID **USC838112TIA**. The primary domain of training and project delivery was Python Programming.

Academic Institution	IILM University, Greater Noida
Degree & Branch	B.Tech Computer Science and Engineering (CSE)
Student Details	Prince Maurya | Roll No: 25SCS1003001538 | Sec: 2CSE7
Internship Role	Python Developer Intern
Industry Partners	upSkill Campus & UniConverge Technologies Pvt. Ltd. (UCT)
Government Partner	AICTE (All India Council for Technical Education)
Certificate ID	USC838112TIA
Duration	12-08-2026 to 02-09-2026
Project Title	URL Shortener Application (GUI + Data Persistence)

2. ORGANIZATION & PROGRAM PROFILE

2.1 UniConverge Technologies Pvt. Ltd. (UCT)

UniConverge Technologies (UCT) is an industry leader in Digital Transformation, specializing in Industrial IoT (IIoT), Cloud Computing, Embedded Systems, and Machine Learning solutions. UCT builds robust products like UCT Insight and Factory Watch, driving industrial automation and sustainable smart factory ecosystems.

2.2 upSkill Campus & The IoT Academy

upSkill Campus is an advanced career development and upskilling platform providing executive mentoring, self-paced technical modules, and real industry project exposure. In partnership with

AICTE, upSkill Campus bridges the gap between college curricula and core industrial requirements.

3. INTERNSHIP OBJECTIVES

- Strengthen Python programming fundamentals, OOP concepts, data structures, and modular code architecture.
- Understand Event-Driven Programming by transitioning from terminal scripts to graphical applications.
- Master local file I/O operations and JSON serialization to achieve data persistence across application reboots.
- Implement robust input validation, exception handling (`try-except`), and error prevention mechanisms.
- Build a complete, user-friendly Desktop URL Shortener Application with clipboard integration.
- Manage code repositories using Git and publish project artifacts on GitHub.

4. LEARNING PROGRESS & WEEKLY MILESTONES

Timeline	Key Milestones & Technical Achievements
Week 01	Project track selection (URL Shortener). Built foundational CLI logic in Python utilizing `string` and `random` libraries to generate 6-character short codes with dictionary mapping.
Week 02	Upgraded from in-memory storage to persistent JSON storage using `json.dump()` and `json.load()`. Added URL validation (`http://`, `https://`) and an interactive CLI loop.
Week 03	Transitioned from CLI to Graphical User Interface (GUI) using `tkinter`. Created window elements, input boxes, buttons, and system clipboard integration (`root.clipboard_append`).
Week 04	UI polish, edge-case validation, exception handling with `messagebox` alerts, "Clear All" functionality, complete end-to-end testing, GitHub documentation, and final report submission.

5. PROJECT DESCRIPTION - URL SHORTENER

The **URL Shortener Application** is a Python desktop software designed to compress long, complex web addresses into clean, 6-character alphanumeric links. Long URLs are often cumbersome to share in documents, emails, or messages. Commercial services like Bitly require cloud subscriptions and online tracking; this project offers a 100% offline, private, lightweight desktop alternative.

The system stores mappings locally in a JSON database (`url_database.json`), ensuring short links persist indefinitely without cloud dependency.

6. PROJECT FEATURES & MODULES

- **URL Shortening Engine:** Accepts long URLs, verifies prefix validity, and generates unique 6-character short links.
- **Collision Prevention Loop:** Checks newly generated short codes against existing database keys to prevent duplicates.
- **Original Link Retrieval:** Accepts a short code or shortened URL and returns the original destination link.
- **System Clipboard Copy:** Allows single-click copying of generated short links directly to the OS clipboard.
- **Persistent Local Storage:** Automatically reads from and writes to a local JSON file on startup and updates.
- **Clear Interface Action:** Resets all entry fields and status messages with a single click.

7. TECHNOLOGY STACK

Technology / Library	Purpose & Role in Project
Python 3.x	Core programming language and application logic.
Tkinter	Standard Python GUI toolkit for building window layout, buttons, and entry fields.
JSON Module	Handles data serialization to read and write database records permanently.
OS Module	Manages file system checks (`os.path.exists`) to avoid missing-file crashes.

String & Random	Generates 6-character alphanumeric combinations (`ascii_letters` + `digits`).
Git & GitHub	Version control and online portfolio hosting.

8. SYSTEM ARCHITECTURE & WORKING

The application follows a modular tier architecture:

1. **User Interface Tier (Tkinter GUI):** Captures user input (URLs or Short Codes) and triggers event commands.
2. **Validation Tier:** Checks for empty inputs and valid HTTP/HTTPS protocol headers.
3. **Algorithmic Execution Tier:** Generates random 6-character strings and executes collision-check loops.
4. **Storage Tier (JSON Database):** Reads and writes string mappings to `url_database.json`.
5. **Clipboard & Feedback Tier:** Updates GUI labels, triggers pop-up alerts, and updates system clipboard memory.

9. DATABASE DESIGN

Data persistence is managed via JSON object key-value pairs stored in `url_database.json`. The short code acts as the primary key mapped to the original target URL.

```
{
    "aB9xK1": "https://www.iilm.edu/academics/computer-science",
    "7mP2qL": "https://github.com/Princemau25/url-shortener",
    "xY3z8N": "https://aicte-india.org"
}
```

10. IMPLEMENTATION DETAILS

The complete backend logic is encapsulated in a clean Python script. Key functions include:

- `load_database()` & `save_database()`: Safe JSON read/write handlers wrapped in `try-except` blocks.
- `is_valid_url(url)`: String validator checking for `http://` or `https://` prefixes.
- `shorten_url()`: Generates short keys, checks for collisions, saves to JSON, and updates GUI state.

- `retrieve_url()`: Splits short URLs to extract the 6-character key and queries the database dictionary.
- `copy_clipboard()`: Invokes `root.clipboard_clear()` and `root.clipboard_append()` with explicit screen updates.

13. TESTING & RESULTS

Test Case Scenario	Input Action	Expected & Observed Result	Status
Empty Long URL	Click 'Shorten' with blank field	Pop-up warning: "Please enter a URL to shorten."	**PASS**
Invalid URL Prefix	Enter "google.com" without http/https	Pop-up warning: "Please include http:// or https://"	**PASS**
Valid Shortening	Enter valid long URL and submit	Generates 6-char short link, updates JSON file.	**PASS**
Clipboard Copy	Click 'Copy to Clipboard'	Short link copied to OS memory; notification shown.	**PASS**
Database Survival	Restart application after shortening	Previously generated URLs successfully retrieved.	**PASS**

14. GITHUB SUBMISSION

The complete source code, JSON sample file, and documentation are published on GitHub:

GitHub Repository: `https://github.com/Princemau25/url-shortener`

Repository Naming Format: `Session_Name_RollNo_Semester_Section`

15. SKILLS & LEARNING OUTCOMES

- Applied Python GUI development skills using Tkinter event loops and widgets.
- Mastered data persistence techniques using JSON serialization and OS file path handling.
- Understood the Software Development Life Cycle (SDLC) through 4 weekly structured sprints.
- Developed defensive programming habits with rigorous input validation and exception handling.
- Enhanced technical writing, documentation, and repository management skills.

16. CHALLENGES & FUTURE SCOPE

Challenges Overcome: Handling clipboard state synchronization across different OS platforms and managing event layout placement in Tkinter.

Future Work Scope:

- Migrate local JSON database to cloud NoSQL storage (e.g., Firebase, MongoDB).
- Convert Tkinter desktop application into a web app using Flask / FastAPI.
- Add user analytics (click counter, geographical access metrics, expiration dates).

17. CONCLUSION & REFERENCES

The Python internship provided invaluable practical exposure, bridging classroom concepts at IILM University with industry software practices at upSkill Campus and UCT. Building the URL Shortener Application consolidated programming logic, GUI development, file persistence, and testing methodologies into a cohesive portfolio project.

References

1. upSkill Campus & UniConverge Technologies (UCT) Internship Program Guidelines (2026).
2. AICTE Virtual Internship Domain Documentation - Python Programming.
3. Python Standard Library Documentation: Tkinter, JSON, Random, and String modules.
4. Official Internship Completion Certificate - Certificate ID: USC838112TIA.

ANNEXURE I: INTERNSHIP COMPLETION CERTIFICATE

INTERNSHIP COMPLETION CERTIFICATE

Issued by upSkill Campus, UniConverge Technologies (UCT) & AICTE

Certificate ID: USC838112TIA

Awarded To: Prince Maurya

Internship Domain: Python

Internship Duration: 12-08-2026 to 02-09-2026

Collaborating Bodies: All India Council for Technical Education (AICTE), UniConverge Technologies Pvt. Ltd., and upSkill Campus.

(Official certificate verified and attached to academic submission)